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A multi-tier methodology for the estimation of individual-specific parameters of DEB models

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ABSTRACT

Intraspecific variability plays a key role in modeling population dynamics and ecological interactions. It is also a fundamental issue for agriculture, livestock production, and aquaculture. Dynamic Energy Budget (DEB) theory is a metabolic theory that can provide insights for these applications, as it can accurately predict the response of an organism to changes in environmental conditions. However, DEB model parameters are typically estimated for the species using mean trait values, neglecting individual differences. With advancing data collection technologies, there is a growing need to leverage individual data to calibrate DEB models. In this paper, we propose a multi-tier methodology for estimating individual-specific DEB parameters based on techniques and principles of the standard DEB estimation procedure. Individual data is often not sufficiently complete, therefore we compliment it available data on organisms of the same species. To avoid estimation bias, we employ a hierarchical tier structure. Data is categorized into tiers, depending on how general it is. In each tier, only a subset of the parameters is estimated depending on available data. The estimation proceeds in a top-down approach, starting with the most general tier and culminating in the individual tier. Values obtained in previously estimated tiers are used as pseudo-data to anchor the parameter estimation. In this way, we obtain biologically sensible individual parameter estimates in an efficient manner. As a case study, we apply our methodology to a dataset that contains weight and feed intake measurements for 52 individuals of the Mertolenga cattle breed from two performance trials. We conclude that variability in the surface-specific maximum assimilation rate and the digestion efficiency is needed to accurately model the individual data. Furthermore, the analysis of tier parameters and individual parameter distributions uncovers differences in the diets fed to the animals.

1. Introduction

Individuals of the same species are similar, but never equal. Intraspecific variation in observed traits stems from both heritable differences and phenotypic plasticity to changing environmental conditions such as non-identical temperature levels or food availability (Moran et al., 2016). It is a key characteristic of populations and it is fundamental to understand population dynamics and interactions with other species, particularly under environmental shifts (Moran et al., 2016; Des Roches et al., 2018; Forsman and Wennersten, 2016). In the field of ecotoxicology, individual variability complicates the determination of the effect of a substance, to the extent that the mean response contains little useful information (Jager, 2013; Augustine et al., 2012). In agriculture, livestock production, and aquaculture, individual variability is a crucial factor where animals with more desirable characteristics are

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actively selected for through breeding programs (Dekkers and Hospital, 2002; Notter, 1999; Abdelrahman et al., 2017).

Dynamic Energy Budget (DEB) theory is a process-based metabolic theory that describes how an organism utilizes the energy in substrate for its maintenance, growth, maturation, and reproduction (Kooijman, 2010; Sousa et al., 2008). DEB theory respects principles of mass and energy conservation and can describe how an organism reacts to changes in environmental conditions such as temperature or food availability, making it an appropriate choice for the underlying organism model of population models (Sousa et al., 2006; Martin et al., 2012; Desforges et al., 2019; Grossowicz et al., 2017). Although DEB theory was formulated from the perspective of the individual organism, due to the lack of available data, DEB model parameters are estimated using mean values of observed traits for the population (Marques et al., 2018). The assumption here is that interindividual variation is small and, when present in datasets, is assumed to be part of the uncertainty or noise of the data (Jager, 2013; Koch and De Schamphelaere, 2020; Potter et al., 2021). In most works that combine DEB and population models, the variability of parameters is ignored or assumed to be a default value given by a rule of thumb (Koch and De Schamphelaere, 2020; Martin et al., 2012).

With advances in sensing technology, gathering data at the individual level is becoming increasingly convenient and prevalent, which highlights the need to estimate individual-specific parameters of DEB models. Recently, several works that leverage this data have been published or added to the Add-my-Pet collection, a database with DEB model parameters for more than 4000 species (Marques et al., 2018). In these works, only a subset of parameters is estimated at the individual or population level, due to a lack of more comprehensible data. They can be categorized into two different types: covariation-based approaches and Bayesian approaches.

Covariation-based approaches estimate a subset of the parameters of the DEB model from the data available for the individual or population by applying the standard DEB estimation procedure, i.e. the covariation method of Lika et al. (2011) with the recent advances in the estimation procedure of Marques et al. (2019). Typically, these approaches do not implement any regularization on the parameter values to constrain the estimation; not even pseudo-data, which is a key characteristic of the covariation method. This can yield unrealistic parameter estimates when the individual data is not sufficient to accurately identify the parameters (Koch and De Schamphelaere, 2020; Potter et al., 2021; Monaco et al., 2019). The exception is the augmented loss function method of Lika et al. (2020), which introduces a regularization term to minimize the squared coefficient of variation of the parameters that are independent for each species in the estimation.

Bayesian approaches allow for the incorporation of uncertainty in measurements and model parameters by assuming a prior distribution (Johnson et al., 2013; Boersch-Supan and Johnson, 2019). The posterior distribution is constrained by the prior distribution and can then be used to derive confidence intervals for any model output, including trajectories (Johnson et al., 2013; Boersch-Supan and Johnson, 2019). However, the estimation procedure is computationally intensive (Boersch-Supan et al., 2017; Boersch-Supan and Johnson, 2019) and if individual parameters are included in the estimation, the size of the parameter space can rapidly increase even with few individuals, intensifying the problem.

In this paper, we present a multi-tier methodology for estimating individual-specific parameters that is an extension of the standard methodology used to estimate parameters of DEB models. First, we categorize the available data into tiers depending on its statistical universe, that is, depending on whether it was gathered at the individual level or from groups of individuals such as populations, breeds, or species. The parameter estimation proceeds in a top-down approach, starting with the most general tier, which is usually the species, and culminating in the individual tier where individual parameters are estimated. In the estimation of each tier, we only include the datasets pertaining to that tier and tiers below. Depending on the data used, only a subset of the parameters is estimated, with the remaining being fixed and equal to the values obtained in the higher tiers. To anchor the estimation of the parameters being estimated, we use pseudo-data with values equal to those estimated in the above tier. In this way, we incorporate into the estimation of individual parameters information that is not present in the individual data, thereby producing realistic parameter estimates, whilst still allowing the optimization to find parameter values that minimize the prediction error. We can parameterize individuals one at a time, meaning that our method is extremely efficient and avoids the high-dimensional search space of methods that estimate parameters for all individuals at once.

As a case study, we apply our methodology to estimate parameters for 52 individuals from the Mertolenga cattle breed (*Bos taurus* Mertolenga), a native breed from the south of Portugal. The estimation data comprises average breed values for several traits, along with results from two performance trials. In these trials, each animal's weight was measured every three weeks and their feed intake was recorded daily. We consider three tiers: breed, trial, and individual. We parameterize a DEB model for the breed and compare several choices of trial and individual parameters.

The remainder of this paper is structured as follows: Section 2 will detail our multi-tier methodology for estimating individual parameters. In Section 3, we describe the available data for the case study, modeling considerations for Mertolenga cattle, exemplify how to apply the methodology, and address some implementation details. Section 4 presents results for each tier considered in the methodology under different hypotheses of trial and individual parameters and discusses the final choice of individual parameters. In Section 5, we provide concluding remarks and outline future work.

2. The multi-tier methodology

2.1. The DEB estimation procedure

DEB models provide predictions for quantities as distinct as weight, reproductive output, or oxygen consumption. Theory interconnects all these quantities using the same set of parameters, with different values for each species. The estimation of parameter values from data is not trivial; since a parameters can appear in the prediction equation for multiple datasets, it is not possible to estimate them separately (Lika et al., 2011; Marques et al., 2019).

The parameter estimation problem is formulated as a minimization problem: the objective is to find the set of parameters $\theta = \{\theta_1, \dots, \theta_{|\theta|}\}$, where $|\theta|$ is the number of parameters, that minimize the deviation between a collection of data $\mathcal{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_{|\mathcal{D}|}\}$, where $|\mathcal{D}|$ is the number of datasets, and the model's predictions for each available dataset $\mathcal{D}_i$. To handle the different units of the datasets and ensure multiplicative symmetry between data and predictions, the loss function in Eq. (1) was proposed by Marques et al. (2019):

$$\underset{\theta}{\text{minimize}} \quad \sum_{i=1}^{|\mathcal{D}|} \sum_{j=1}^{|\mathcal{D}_i|} \frac{w_{ij}}{|\mathcal{D}_i|} \frac{\left(d_{ij} - p_{ij}\right)^2}{\bar{d}_i^2 + \bar{p}_i^2}, \tag{1}$$

$$\text{with} \quad \bar{d}_i = \frac{1}{|\mathcal{D}_i|} \sum_{j=1}^{|\mathcal{D}_i|} d_{ij}, \quad \bar{p}_i = \frac{1}{|\mathcal{D}_i|} \sum_{j=1}^{|\mathcal{D}_i|} p_{ij} \quad \text{and} \quad p_{ij} = p_{ij}(\theta),$$

where $|\mathcal{D}_i|$ is the number of datapoints in dataset $\mathcal{D}_i$, d_{ij} is the value of datapoint j of dataset $\mathcal{D}_i$, and p_{ij} is its corresponding prediction which is a function of the set of parameters θ being estimated. The weight w_{ij} quantifies the relative reliability of datapoint j of dataset $\mathcal{D}_i$ and is typically set to 1. Dividing by the number of datapoints $|\mathcal{D}_i|$ is used to ensure that each dataset contributes equally to the loss function (Marques et al., 2018, 2019). This number also defines an important distinction: that of zero-variate and multivariate datasets. A zero-variate dataset consists of a single datapoint ($|\mathcal{D}_i| = 1$), e.g. maximum weight or maximum reproduction rate. Conversely, multivariate

datasets have one or more independent variables and a single dependent variable (Lika et al., 2011). Length as a function of age or oxygen consumption as a function of weight and temperature are examples of multivariate data.

The set of parameters θ that are estimated does not include all parameters in DEB theory, it typically only includes the 12 core primary parameters (Kooijman, 2010; Lika et al., 2011). The auxiliary parameters, related to temperature and chemical properties, are usually fixed with standard values being used for most organisms. This is because a parameter should only be estimated if it is identifiable from the available data types. For example, the Arrhenius temperature T_A can only be estimated if data at several temperature levels is available, as other types of data do not provide any information on the values of this parameter.

The parameters that minimize the loss function are not necessarily the best estimates for the species. If the data is not sufficiently complete or if it is of low quality, the estimation of some parameters can be ill-defined and there may exist several possible parameter values (or even an infinite number) that achieve similar loss function values. To address this issue and to further increase identifiability, pseudo-data was introduced. Pseudo-data consists of values for several parameters (or functions of parameters) for a generalized animal and were determined based on existing estimates for species in the Add-my-Pet collection (Kooijman, 2010; Lika et al., 2011).

Pseudo-data values are treated in the same way as data in the loss function, although with lower weights so as to not excessively control the parameter estimation (Marques et al., 2019; Lika et al., 2011). With the addition of pseudo-data, the loss function now becomes:

$$\underset{\theta}{\text{minimize}} \quad \sum_{i=1}^{|\mathcal{D}|} \sum_{j=1}^{|\mathcal{D}_i|} \frac{w_{ij}}{|\mathcal{D}_i|} \frac{\left(d_{ij} - p_{ij}\right)^2}{\bar{d}_i^2 + \bar{p}_i^2} + \sum_{k=1}^{|\pi|} w_k \frac{(\theta_k - \pi_k)^2}{\theta_k^2 + \pi_k^2}, \tag{2}$$

where $\pi = \{\pi_1, \dots, \pi_{|\pi|}\}$ is the set of pseudo-data values with size $|\pi|$, π_k is the pseudo-data value for parameter θ_k being estimated, and w_k is its weight value. We see the use of pseudo-data as a form of regularization, a technique widely used in statistics and machine-learning that introduces penalties on parameter values during estimation (Bickel et al., 2006; Girosi et al., 1995). These penalties help yield parameter estimates that do not overfit to the available data. In addition to making more parameters identifiable, pseudo-data regularizes parameter estimates to align with existing estimates in the Add-my-Pet collection.

To minimize the loss function, usually the Nelder–Mead simplex algorithm is applied (Nelder and Mead, 1965; Robles et al., 2023). The Nelder–Mead algorithm is a robust algorithm that has been widely employed in several scientific areas; however, there are no guarantees of convergence and its performance has been shown to deteriorate as the dimensionality increases (Han and Neumann, 2006).

Finally, goodness-of-fit should not be the only criterion when analyzing the results of an estimation. The estimation should be done in-context by analyzing model predictions for biological reasonableness and by comparing parameter values with those of taxonomically similar species. For this reason, parameter estimation of DEB models is often an iterative process requiring the rethinking of assumptions and adjustments of the weights of each dataset and each pseudo-data value.

2.2. On the estimation of individual parameters

DEB theory describes the metabolism of an individual organism. It incorporates a set of assumptions, such as the choice of state variables or weak and strong homeostasis, which together with mass and energy balances specify the form of the model (Sousa et al., 2008; Kooijman, 2010). The parameters that emerge from theory are assumed to vary both between and within species. From this assumption, it follows that individual data should provide the most accurate parameter estimates for the DEB model of a given organism. The accurate estimation of all model parameters for a single organism requires a high degree of data completeness, according to the scale of Lika et al. (2011). Achieving high completeness is, however, not easy; to date, the species with highest completeness in the Add-my-Pet collection is *Daphnia magna* and it only achieves a 6/10 in the completeness scale.

The reason for this is that DEB models require comprehensive data on an organism. Most parameters are only identifiable if data is available at several food levels. Estimating chemical composition parameters requires measurements of elemental composition and elemental balances. Parameters related with aging and maximum size require measurements throughout the life of an organism, which is particularly challenging for those with longer lifespan (Lika et al., 2011). Therefore, experimental designs capable of gathering individual data with high completeness are likely very expensive and intractable for most species and researchers. The consequence is that most parameters will usually not be identifiable from individual data alone.

Given the challenge of acquiring individual data, DEB models are rarely calibrated for the individual. Instead, a single DEB model is calibrated for the species as a whole using mean values for several traits. We refer to these datasets as group data, as the values were obtained from a group of individuals, as opposed to individual data where the data is gathered from a single organism. By including group data in the estimation and only estimating a single set of species parameters θ, the effect of intraspecific variability of parameters on trait measurements is assumed to be part of the uncertainty or noise (Jager, 2013; Koch and De Schamphelaere, 2020; Potter et al., 2021). This has been the approach for most of the Add-my-Pet collection, which contains parameter estimates for thousands of species (Marques et al., 2018).

Since group data is more easily obtainable, we can combine it with individual data in the estimation to render more parameters identifiable at the individual level. Let us formalize this distinction in the nature of the dataset, regardless of whether the dataset is zero-variate or multivariate:

- *Individual data* – Data gathered at the individual level. Each dataset of individual data $\mathcal{I}_{li}$ is composed of $|\mathcal{I}_{li}|$ datapoints d_{lij} measured from the same individual l.
- *Group data* – Data obtained from a group of individuals. Each dataset of group data $\mathcal{G}_{gi}$ is composed of $|\mathcal{G}_{gi}|$ datapoints d_{gij} measured from a sample of individuals that belong to the same statistical universe g.

The statistical universe in group data can refer to a species or a population of a species. For example, data on the mean lifespan of a species is obtained from a sample of individuals that are a subset of all individuals of the species, which means that the statistical universe is the entire species. Or, if we have a dataset of individual oxygen consumption for a population of fish that inhabit a lake, then these individuals constitute a sample of the population of that lake.

Sometimes, group data values can be obtained without a sample being explicitly defined. Values for specific traits of species, breeds or populations can be found in the literature or provided by experts where a rigorous experimental design was not considered or not reported. We refer to these datasets as expert knowledge datasets. They typically consist of single values (zero-variate data) and are not respective to a single individual, therefore we consider them to be group data. Even though these values are not obtained from a concrete sample, conceptually, these datasets are similar to a mean value calculated based on a sample of individuals from the same statistical universe.

Considering this distinction, let us formulate the individual parameter estimation problem when a collection of individual data $\mathcal{I}$ and a collection of group data $\mathcal{G}$ are available. Now, the objective is to find a set of parameters $\theta_l = \{\theta_{l1}, \dots, \theta_{l|\theta_l|}\}$ for each individual l, such that we minimize the deviation between the predictions and the individual data $\mathcal{I} = \{\mathcal{I}_1, \dots, \mathcal{I}_{|\mathcal{I}|}\}$ of each individual l and the group data $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_{|\mathcal{G}|}\}$ from each statistical universe g. The set of parameters of each individual θ_l includes all parameters that are identifiable from

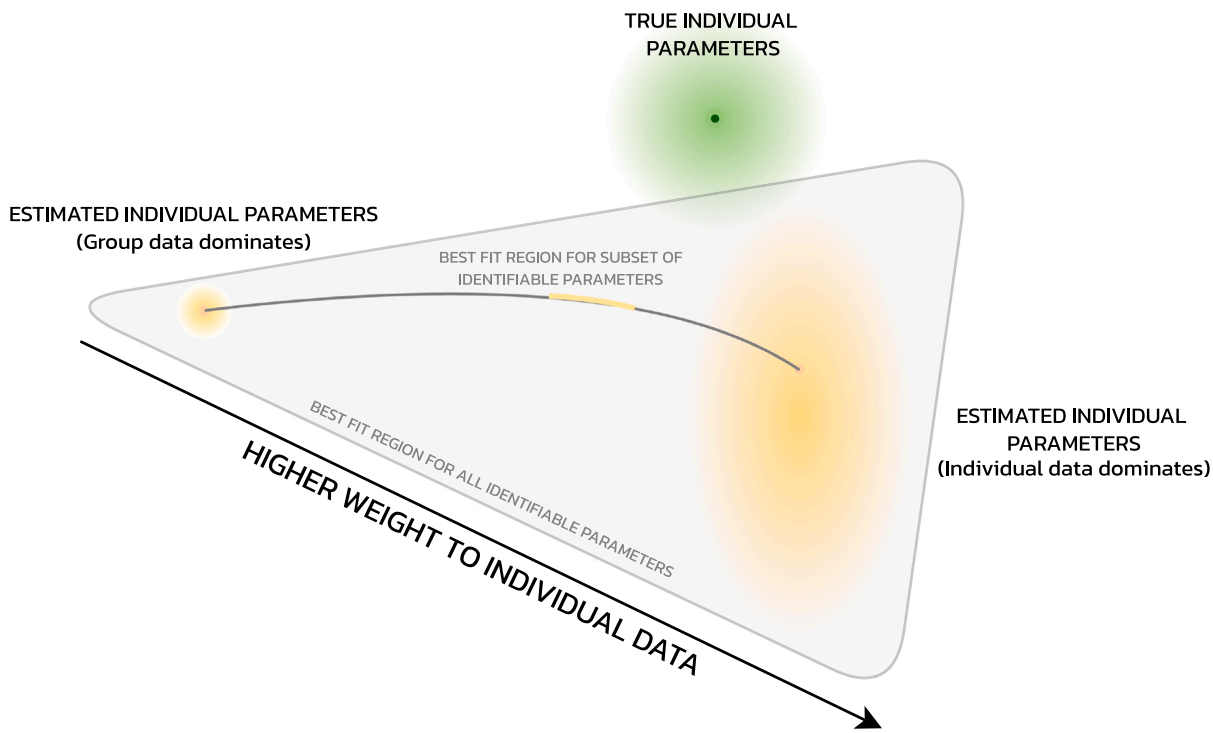

Fig. 1. The trade-off between bias and variance in the estimation of individual parameters.

both group and individual data. The loss function is similar to the standard loss function of Eq. (1), but with a term for individual data and a term for group data:

$$\underset{\theta_1,\ldots,\theta_l,\ldots,\theta_{|\mathcal{I}|}}{\text{minimize}} \quad \sum_{l=1}^{|\mathcal{I}|}\sum_{i=1}^{|\mathcal{I}_l|}\sum_{j=1}^{|\mathcal{I}_{li}|} \frac{w_{lij}}{|\mathcal{I}_{li}|}\frac{\left(d_{lij}-p_{lij}\right)^2}{\bar{d}_{li}^2+\bar{p}_{li}^2} + \sum_{g=1}^{|\mathcal{G}|}\sum_{i=1}^{|\mathcal{G}_g|}\sum_{j=1}^{|\mathcal{G}_{gi}|} \frac{w_{gij}}{|\mathcal{G}_{gi}|}\frac{\left(d_{gij}-p_{gij}\right)^2}{\bar{d}_{gi}^2+\bar{p}_{gi}^2}, \quad (3)$$

$$\text{with} \quad \bar{d}_{li} = \frac{1}{|\mathcal{I}_{li}|}\sum_{j=1}^{|\mathcal{I}_{li}|} d_{lij}, \quad \bar{p}_{li} = \frac{1}{|\mathcal{I}_{li}|}\sum_{j=1}^{|\mathcal{I}_{li}|} p_{lij}, \quad p_{lij} = p_{lij}(\theta_l),$$

$$\bar{d}_{gi} = \frac{1}{|\mathcal{G}_{gi}|}\sum_{j=1}^{|\mathcal{G}_{gi}|} d_{gij}, \quad \bar{p}_{gi} = \frac{1}{|\mathcal{G}_{gi}|}\sum_{j=1}^{|\mathcal{G}_{gi}|} p_{gij}, \quad p_{gij} = p_{gij}(\theta_1,\ldots,\theta_{|\mathcal{I}|}),$$

where $|\mathcal{I}|$ is the number of individuals, $|\mathcal{I}_l|$ is the number of datasets of individual l, $|\mathcal{I}_{li}|$ is the number of datapoints in dataset $\mathcal{I}_{li}$ of individual l, $|\mathcal{G}|$ is the number of statistical universes, $|\mathcal{G}_g|$ is the number of datasets of statistical universe g, and $|\mathcal{G}_{gi}|$ is the number of datapoints in dataset $\mathcal{G}_{gi}$ of statistical universe g. The predictions p_{lij} for individual datapoints d_{lij} are functions of the individual parameters θ_l of individual l.

The predictions p_{gij} for group datapoints d_{gij} should be functions of the individual parameters θ_l of all individuals, as the data is composed of mean values. The form of the function p_{gij} is arbitrary, but it should be an average of the predictions for each datapoint p_{gij} using the parameters of each individual θ_l, instead of using the average of the parameters. The reasoning for this follows from two facts: DEB theory is formulated for the organism and the equations for most traits are nonlinear. Because the measurements are mean values, they will not be accurately described by the equation. The nonlinearities mean that a single prediction using average parameter values will result in an erroneous estimate of the trait. When we average the predictions, this source of error is eliminated; however, to compute the average, we need to predict group data using the parameters of each individual. If we have many individuals, then the estimation problem can even become intractable to solve due to the large number of computations required.

Another issue with the use of group data is that it introduces bias in the estimation of individual parameters. By bias, we mean the difference between the parameter estimates we obtain after minimizing the loss function and the correct parameters. Bias will always exist in parameter estimation; however, modelers should strive to reduce it as much as possible.

We can identify two sources of bias from group data: environmental bias and sample bias. Environmental bias occurs when the individuals in our estimation did not experience the same environmental conditions of the organisms of the statistical universe of a group dataset. These conditions include, for example, food availability, food quality, and temperature. When these environmental differences are not accounted for, or are impossible to account for, what would be a difference in environmental forcing variables is converted during estimation to a difference in parameter values.

Sample bias arises from the fact that all organisms are distinct from each other. It occurs if the group of individuals of a group dataset are not similar to the individuals in the estimation or if the individuals in the estimation are not part of the statistical universe of a group dataset.

Since group data is not specific for an individual and because bias will naturally exist due to the use of group data, achieving zero deviation in the predictions for group data is not necessarily the goal. Given that the traits of an individual will likely not lie on the mean, due to genotypic and phenotypic differences, there will be an acceptable deviation. How much deviation is acceptable, however, is difficult to determine without measures of dispersion. Furthermore, during estimation, the deviation between group data and predictions will be a combination of the acceptable deviation and the error from inaccurate parameter estimates. When measures of dispersion are available, these should be integrated in the estimation and it is up to the modeler to decide on the acceptable deviation.

During estimation, individual data and group data will guide the parameters towards different regions of the parameter space. How much group data controls the estimation, compared to individual data, is given by their relative weights. The higher the importance given to

group data, the more biased the individual parameter estimates will be, for the reasons outlined before. If instead individual data is given higher importance, the estimation will have higher variance.

In this situation, variance refers to how dissimilar the parameters of each individual are from each other. Since the estimation is on the set of parameters θ_l that is identifiable from both group and individual data, the model will have more degrees of freedom than required to achieve a good fit on individual data. When individual data is prioritized, these degrees of freedom will cause the model to overfit on individual data without considering other aspects of the metabolism. As a result, the parameter estimates of each individual will be very different from each other and the estimation will have high variance. Consequently, these estimates can end up being very far from the best parameters for the individual whilst still minimizing the loss function.

By including group data in the estimation of individual parameters, we have solved the identifiability problem. The consequence is that we have introduced a bias–variance trade-off, and we must find a balance in order to obtain the best parameter estimates. The trade-off is exemplified in Fig. 1. The true parameters of each individual are represented by the green distribution. The gray region represents the region of the DEB model parameter space with the best fit for each individual, depending on the relative weights of group and individual data. The gray region does not fully enclose the green parameter distribution because we have fixed unidentifiable parameters; the fixed parameter will likely not have the true values for the individual, which introduces some unavoidable bias in the estimation. When group data is given higher weight, the individual parameter distribution is less sparse (low variance), but in general the parameters are further away from the true values (high bias). When individual data dominates the estimation, the parameters are on average closer to the true values (low bias), but the distribution is more sparse (high variance) and as a result for some individuals the estimated parameters are far from the true values. The next section describes techniques to handle this trade-off and how to structure the estimation in order to further reduce bias.

2.3. Reducing bias and variance with estimation tiers

To handle the bias–variance trade-off, we must reduce the bias from group data whilst also reducing the variance in the estimation of individual parameters.

To reduce variance, we start by reducing model complexity by only estimating a subset of the model parameters for each individual ϕ_l that are identifiable only from individual data. This set of parameters is a subset of all parameters θ that are identifiable from both group and individual data. This is the approach taken by most works that deal with individual parameters of DEB models (Koch and De Schamphelaere, 2020; Potter et al., 2021; Monaco et al., 2019).

The parameters that do not appear in the set ϕ_l are still identifiable from both group and individual data. Instead of estimating them for each individual, we now estimate a single set of parameters θ that is shared by all individuals. The set of parameters θ can be thought of as a set of species parameters, similarly to the estimation problem defined in Eq. (2).

By not estimating all identifiable parameters at the individual level, we are introducing bias in the estimation, as in reality the parameters of each individual will not have a value equal to the estimated species parameter. As a trade-off we reduce variance, which will likely let us obtain, in general, parameter estimates closer to the true values. In Fig. 1, the gray line represents the parameter space where only ϕ_l is estimated for each individual, with the yellow segment representing the variance. Since the individual parameters vary within a lower-dimensional region of the parameter space, the overall variance of the estimation is reduced.

To further reduce variance, we take inspiration from pseudo-data by introducing a regularizer on individual parameter values to minimize their deviation from the species parameters θ. This means that the parameters that appear in ϕ_l are estimated for each individual and also for the species. This regularizer ensures that the parameters of each individual do not deviate significantly from the species parameters which could result in biologically unreasonable predictions.

Using these two strategies to reduce variance, we can combine the loss functions of Eqs. (2) and (3) into the following loss function:

$$\underset{\theta,\phi_1,\ldots,\phi_l,\ldots,\phi_{|\mathcal{I}|}}{\text{minimize}} \quad \sum_{l=1}^{|\mathcal{I}|}\sum_{i=1}^{|\mathcal{I}_l|}\sum_{j=1}^{|\mathcal{I}_{li}|}\frac{w_{lij}}{|\mathcal{I}_{li}|}\frac{\left(d_{lij}-p_{lij}\right)^2}{\bar{d}_{li}^2+\bar{p}_{li}^2}+\sum_{g=1}^{|\mathcal{G}|}\sum_{i=1}^{|\mathcal{G}_g|}\sum_{j=1}^{|\mathcal{G}_{gi}|}\frac{w_{gij}}{|\mathcal{G}_{gi}|}\frac{\left(d_{gij}-p_{gij}\right)^2}{\bar{d}_{gi}^2+\bar{p}_{gi}^2}+ \tag{4}$$

$$+\sum_{k=1}^{|\pi|}w_k\frac{(\theta_k-\pi_k)^2}{\theta_k^2+\pi_k^2}+\sum_{l=1}^{|\mathcal{I}|}\sum_{k=1}^{|\phi_l|}w_k\frac{(\phi_{lk}-\theta_k)^2}{\phi_{lk}^2+\theta_k^2},$$

$$\text{with} \quad p_{lij}=p_{lij}(\theta,\phi_l) \quad \text{and} \quad p_{gij}=p_{gij}(\theta),$$

where the third term in the loss function is the loss for the pseudo-data of the species parameters θ, and the fourth term is the regularizer on the individual parameters ϕ_l that penalizes deviations from the species parameters. The predictions for the individual data of individual l are calculated with its individual parameters ϕ_l and the species parameters θ.

The predictions for group data are now computed only with the species parameters. This helps reduce the bias from group data on the estimation of individual parameters, as the individual parameters are not directly determined from group data. However, since the predictions for individual data use both individual parameters ϕ_l and species parameters θ, they are still influenced by the group data and bias will still exist.

Furthermore, because the estimation of individual parameters is done in conjunction with the estimation of species parameters, the species parameters can also be biased if the individuals are not a representative sample of the species or if the environmental conditions are not taken into account. Essentially, the sources of bias from group data described in Section 2.2 also occur for individual data with regards to the estimation of species parameters. This situation is aggravated the more weight is given to individual data in the loss function.

To solve this, we propose working in estimation tiers by first estimating a set of species parameters θ from all data and then estimating the set of individual parameters ϕ_l of each individual l using only its own individual data. With estimation tiers, we separate the estimation of individual and species parameters and reduce the bias in the estimation of both. Furthermore, we can use intermediate estimation tiers to obtain a set of parameters with lower bias with respect to the individual parameters.

Estimation tiers are derived from the statistical universes of available data. The universes of all datasets can be related hierarchically: e.g., an individual is a subset of all the individuals of a population, and in turn, the population forms a subset of all individuals of the species. These hierarchical relationships form the basis for the tiers in the estimation procedure, with the individual tier being the final tier.

In each estimation tier, we will estimate a set of tier parameters ϕ_t for every statistical universe t in the tier. For example, in a population tier, we can estimate ϕ_t for two separate populations. The parameters in set ϕ_t are a subset of all identifiable parameters θ. Only group datasets relative to the statistical universe t and all statistical universes that are subsets of it are included in the estimation. Similarly, only datasets of individuals that are part of the statistical universe t are included in the estimation. The parameters in ϕ_t should, therefore, be identifiable based only on the group data collection for the tier $\mathcal{G}^t$ and the individual data collection $\mathcal{I}^t$. Let us define the optimization problem for a tier:

$$\underset{\phi_t}{\text{minimize}} \quad \sum_{l=1}^{|\mathcal{I}^t|}\sum_{i=1}^{|\mathcal{I}_l^t|}\sum_{j=1}^{|\mathcal{I}_{li}^t|}\frac{w_{lij}}{|\mathcal{I}_{li}^t|}\frac{\left(d_{lij}-p_{lij}\right)^2}{\bar{d}_{li}^2+\bar{p}_{li}^2}+\sum_{g=1}^{|\mathcal{G}^t|}\sum_{i=1}^{|\mathcal{G}_g^t|}\sum_{j=1}^{|\mathcal{G}_{gi}^t|}\frac{w_{gij}}{|\mathcal{G}_{gi}^t|}\frac{\left(d_{gij}-p_{gij}\right)^2}{\bar{d}_{gi}^2+\bar{p}_{gi}^2}+ \tag{5}$$

$$\sum_{k=1}^{|\phi_l|}w_k\frac{(\phi_{lk}-\theta_{vk})^2}{\phi_{lk}^2+\theta_{vk}^2}, \quad \text{with} \quad p_{lij}=p_{lij}(\theta_v,\phi_t) \quad \text{and} \quad p_{gij}=p_{gij}(\theta_v,\phi_t),$$

Table 1
Breed data for the Mertolenga breed and corresponding predictions and relative error after the estimation of breed parameters.

Data	Source	Value	Prediction	RE (%)	Unit
Gestation time	Quaresma et al. (2004)	284	276	2.9%	d
Lifespan	Bettencourt (2021)	20	20	0.0%	year
Maximum reproduction rate	ACBM (2023)	0.0022	0.0022	0.2%	calves/d
Time since birth at weaning for females	ACBM (2023)	210	223	6.3%	d
Time since birth at puberty for females	ACBM (2023)	289	314	8.8%	d
Maximum withers height for males	FAO (2023)	155	150	3.1%	cm
Wet weight at birth for females	Quaresma et al. (2004)	46	47	3.2%	kg
Wet weight at weaning for females	Carolino et al. (2020)	155.4	142.5	8.3%	kg
Wet weight at puberty for females	Expert knowledge	205	183	10.7%	kg
Maximum wet weight for females	FAO (2023)	400	447	11.8%	kg
Time since birth at weaning for males	ACBM (2023)	210	212	1.1%	d
Time since birth at puberty for males	Monteiro et al. (2012)	350	384	9.6%	d
Maximum withers height for males	FAO (2023)	165	170	3.2%	cm
Wet weight at birth for males	Quaresma et al. (2004)	46	53	15.8%	kg
Wet weight at weaning for males	Carolino et al. (2020)	177.8	175.4	1.4%	kg
Wet weight at puberty for males	Expert knowledge	320	287	10.3%	kg
Maximum wet weight for males	FAO (2023)	700	730	4.2%	kg

where $|\mathcal{I}^t|$ is the number of individuals and $|\mathcal{G}^t|$ is the number of statistical universes in the tier estimation. The predictions for both group and individual data are functions of the tier parameters ϕ_t and a fixed set of parameters θ_v that has values equal to those estimated in the previous tier v.

The estimation will then proceed in a top-down approach starting with the species tier, going through each intermediate tier, with the final tier being the individual tier. In the species tier, the estimation follows the standard methodology, The tier parameters are the species parameters $\phi_t = \theta$, the set of parameters identifiable from all available data, and pseudo-data is included to constrain their estimation. In the individual tier, the tier parameters are the individual parameters $\phi_t = \phi_I$, estimated only from individual data.

With each estimation tier, we reduce sample bias and environmental bias from group data for the estimation of individual parameters, since the individuals and environmental conditions are increasingly similar to those of the individuals we are estimating for. Given that we are fixing most parameters and regularizing the free ones, the estimation will have reduced variance. In the end, we obtain biologically reasonable parameter estimates that still provide accurate predictions for individual data.

The parameters of each tier are also useful for predictions: they are a centrality measure of the parameter distributions of the individuals that belong to the statistical universe of the tier. For example, if we have a population tier and estimate tier parameters for each population with available data, we can calculate population-specific predictions and draw insights based on the differences between them.

2.4. The selection of tier and individual parameters

In the application of the multi-tier methodology, the most important decision is that of selecting tier parameters for each tier. On the basis of the types of group and individual data in a given tier and all tiers below it, some parameters will not be identifiable. For example, if we do not have feeding data, then the assimilation efficiency κ_X cannot be estimated, and if we have a maximum length for the species, we can estimate a shape factor as a species parameter, but we cannot estimate a shape factor for a population for which we do not have length data.

The characteristics of the data are also relevant for the identifiability of parameters. If the data does not include the full lifecycle of an individual, then the number of degrees of freedom is reduced, and as a consequence we can estimate fewer parameters. Typically, the degrees of freedom of the DEB equation for a given trait are more easily expressed in terms of compound parameters, which are functions of core parameters. For instance, for a von Bertalanffy growth curve we have two degrees of freedom, which are two compound parameters: the von Bertalanffy growth rate $\dot{r}_B$ and the ultimate length L_∞. However, if the growth curve in our data never starts to plateau, then we do not have information on the two compound parameters and can only estimate one of them.

Since in the estimation of tier parameters most parameters are fixed, the estimation of a compound parameter is effectively the same as the estimation of a core parameter. This implies that a choice must be made as to which core parameters will be the tier parameters. This choice should be made based on the parameters that are more likely to vary between individuals. In some cases, the choice may not be obvious, or several plausible explanations may exist. In these cases, we recommend running the multi-tier procedure with several possible choices of tier parameters that are biologically sensible.

The appropriate choice of individual parameters will result in a proper model fit for every type of individual data. However, it is not guaranteed that there will be a single parameter choice that results in a good fit and the final choice is up to the modeler.

2.5. The multi-tier estimation procedure

The first step in applying the multi-tier methodology is to structure the data into tiers based on their statistical universes. Then, the choice of tier parameters must be made taking into account the identifiability of parameters and their biological reasonableness. With these decisions made, the estimation of tier parameters is done sequentially, starting with the highest tier and ending in the individual tier.

In each tier, only the tier parameters are estimated. The remaining parameters are fixed, and have values equal to those estimated in the above tier. The exception is the first tier, where standard values are used for parameters that are unidentifiable, as is usual in the standard DEB estimation procedure.

In the estimation, we include all data pertaining to the tier for which parameters are estimated and tiers below for the reasons highlighted previously in Sections 2.2 and 2.3. We set the weight of each datapoint of a dataset of individual data w_{lij} as

$$w_{lij} = \frac{1}{|\mathcal{I}_i^t|}, \tag{6}$$

where $\mathcal{I}_i^t$ is the number of individuals that have measurements of the type of dataset $\mathcal{D}_i$ and that belong to a sample of the same statistical universe t.

The rationale for this choice is to assign the same total weight regardless of whether the dataset for a given trait is composed of individual data from individuals of the same sample or is a dataset of group data. This avoids situations where the number of individual datasets is significantly larger than the number of datasets of group data and a standard value of $w_{lij} = 1$ would cause individual data to disproportionately influence the minimization of the loss function. For

cases where the number of populations or other groups of individuals is large, we also recommend setting a lower weight. In the individual tier, since only individual data is included, we can set the weight of individual data equal to the standard value.

To anchor the estimation, the values of the tier parameters obtained in the above tier are used both as initial values and as pseudo-data. We set the weight of pseudo-data to the standard value of 0.1. Furthermore, filters on parameter values (Marques et al., 2018) are applied in each tier of the estimation, to ensure that the full set of parameters of each group and individual is biologically consistent.

Following the construction of the estimation files according to these guidelines, the estimation procedure can be run.

In each tier, if there statistical universes with data independent from each other, then their parameters will not influence each other, and the parameter estimation can be run separately for each one. For example, if we have data on the frequency of deaths for several populations, the Gompertz stress coefficient can be estimated for each one. If no plausible reason exists for the deaths of one population to influence those of another, this parameter should be independent for each population, and the estimation can be run separately.

This characteristic is particularly relevant for the individual tier where a large number of individuals can exist and so a parameter space with large dimensionality. Since the parameters of each individual do not influence each other, their estimation is independent and each one can be run individually. This improves performance significantly and avoids large dimensionality parameter spaces where the performance of Nelder–Mead deteriorates. Estimating parameters for a DEB model is often an iterative process, requiring the rethinking of assumptions or adjustments in the weights of each dataset, followed by an interpretation of the results and parameter estimates. By running the estimation of each individual separately, the multi-tier estimation procedure enables faster and easier iteration of the parameter estimation.

3. A case study for Mertolenga cattle

3.1. Available data

For the case study, we collected data on the Mertolenga breed from the literature and obtained individual data on 55 bulls.

Breed data consists of gestation time, lifespan, reproduction rate, maximum length, and age and weight at all life stages (birth, weaning, puberty, and fully grown) for both males and females. Sources and values are provided in Table 1.

Individual data consists of weight and feed intake data collected by the Portuguese association of Mertolenga cattle breeders (ACBM - Associação de Criadores de Bovinos Mertolengos) during productive performance trials that aim to evaluate the productive characteristics and genetic potential of the animals (Harris and Newman, 1994; Raidan et al., 2016). Two trials were carried out in 2020 and 2021. Each animal was housed independently and the trials were performed in the same location, under similar environmental conditions. All animals in the trial were fed the same diet, which was designed to promote high growth rates. While the diets used in each trial were similar in nutritional content, they were not identical.

In total, data comprises 55 bulls with initial ages ranging between 12.4 and 16.4 months. Data on three animals was discarded due to illness, leaving 25 individuals in the 2020 trial and 27 individuals in the 2021 trial. Both tests had a duration of 4 months, started in August, and were preceded by an adaptation period of 3 weeks. Weight measurements were taken every 3 weeks, in the morning, after fasting during the night. Feed intake was measured daily for each individual using a Roughage Intake Control (RIC) machine from Hokofarm (Chapinal et al., 2007). The dry matter content of the feed was also measured several times during the trials.

3.2. A DEB model for mertolenga cattle

An stx model (Marques et al., 2018) was calibrated for the breed *Bos taurus* Mertolenga. The model assumes that fetal development occurs with an initial preparation stage, called diapause, where development begins only at a time t_0. Since no significant growth was observed in bovine fetuses before 80 days (Prior and Laster, 1979), we used this value as pseudo-data to inform the estimation of t_0.

Males and females are assumed to differ in the values of the surface-specific maximum assimilation rate $\{\dot{p}_{Am}\}$ as well as the maturity levels at weaning and puberty, E_H^x and E_H^p. The data value for weaning age is 7 months, although there are studies that suggest that weaning occurs naturally at a later date (Enríquez et al., 2011; Reinhardt and Reinhardt, 1981). We chose to use this value as weight predictions at weaning are of significant importance in production settings and opting for other values would impact the applicability of the model.

The scaled functional response f, which describes the degree of food availability between 0 and 1, was set to its maximum, since cattle are typically fed *ad libitum*. Furthermore, since the animals were housed in individual pens, no food competition occurred. Any differences in food quality are instead incorporated in the differences in the values of $\{\dot{p}_{Am}\}$.

With maximum scaled function response, growth occurs at constant food and the structural length $L(t)$ follows the von Bertalanffy curve:

$$L(t) = L_\infty - (L_\infty - L_i)\exp(-\dot{r}_B t), \tag{7}$$

$$\dot{r}_B = \frac{1}{3}\frac{[\dot{p}_M]}{f\kappa\{\dot{p}_{Am}\}/\dot{v} + [E_G]},$$

where $\dot{r}_B$ is the von Bertalanffy growth rate. L_i is the initial length and L_∞ is the ultimate length:

$$L_\infty = fL_m - L_T = f\frac{\kappa\{\dot{p}_{Am}\}}{[\dot{p}_M]} - \frac{\{\dot{p}_T\}}{[\dot{p}_M]}, \tag{8}$$

with L_m as the maximum length and L_T as the heating length. We consider $\{\dot{p}_T\} = 0$ as the available data does not allow the estimation of this parameter.

At constant food, the reserve density remains constant due to weak homeostasis, which means we can write the weight as:

$$W(t) = (1 + f\omega)L(t)^3, \quad \omega = \frac{w_E[E_m]}{d_E\bar{\mu}_E}, \tag{9}$$

where ω is the contribution of reserve to weight, $[E_m] = \{\dot{p}_{Am}\}/\dot{v}$ is the maximum reserve density, and w_E, d_E, $\bar{\mu}_E$ are, respectively, the molecular weight, the density and the chemical potential of reserve. The daily feed intake, on a dry matter basis, can be calculated from the ingestion power of food $\dot{p}_X$ and converted to mass with the molecular weight w_X and chemical potential $\bar{\mu}_X$ of food:

$$DMI(t) = \frac{w_X}{\bar{\mu}_X}\dot{p}_X = \frac{fw_X\{\dot{p}_{Am}\}}{\bar{\mu}_X\kappa_X}L(t)^2. \tag{10}$$

In the predictions for weight and feed intake, the initial weight was set equal to the initial weight measurement of the individual. The reason behind this choice is that previous life history information about the animals is not available for estimation. If the predictions were to be calculated from birth, the presence of environmental differences would complicate the process of fitting the model. This, in turn, would make it more difficult to distinguish what part of the variability in traits is due to parameter variability and what part is due to varying environmental factors. Additionally, given that all animals undergo an adaptation period, they should be at the same reserve density ($e = f = 1$) when the experiment starts, minimizing the effect of previous environments.

3.3. Applying the methodology

For this case study, we have breed and individual data available. To account for different environmental conditions between the performance trials and the typical conditions of breed data, we will consider

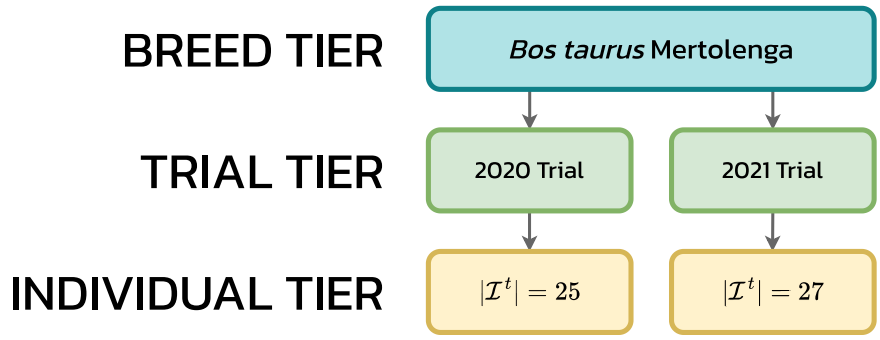

Fig. 2. The tier structure for the case study on Mertolenga cattle.

an intermediate trial tier. As there are two performance trials, we will consider two groups, one for each trial, to model differences in the diets and other environmental factors. In the individual tier, each individual will use the trial parameter associated with the trial it belongs to as pseudo-data. The tier structure is presented in Fig. 2.

In the breed tier, we use both breed and individual data to estimate a set of breed parameters for the Mertolenga breed. The weight of individual data in the breed tier is set according to Eq. (6), depending on the number of individuals that took part in the performance trial of each year. The available data allows the estimation of most parameters typically estimated for DEB models (see Table 3).

Since there is no trial group data, the trial parameters will be estimated using only individual data. Having two types of data, growth and feed intake, we can estimate at most the von Bertalanffy growth rate $\dot{r}_B$, and the maximum size L_m, the maximum reserve density E_m, and the digestion efficiency κ_X. The chemical composition parameters that appear in the growth and feed intake equations are not identifiable based solely on these datasets. Furthermore, they are not likely to vary substantially within a species to explain the variability in the data.

Within the duration of the performance trials, the animals are still growing and do not reach maximum size (see Fig. 7 for an example). Therefore, the von Bertalanffy growth curve is not complete and we can estimate at most two of the four parameters listed above. Of these, the first three are compound parameters and are functions of the following parameters: $\{\dot{p}_{Am}\}$, κ_X, $[\dot{p}_M]$, $\dot{v}$, κ, and $[E_G]$. We can disregard $[E_G]$ because its effect on growth is small due to the large maximum reserve density of large animals. The intraspecific variation in the other parameters is likely to explain the variability in the data for the following reasons:

- $\{\dot{p}_{Am}\}$ – Controls assimilation. Often used to model differences in maximum size.
- κ_X – Connects ingestion to assimilation. Can explain variability in feeding.
- $[\dot{p}_M]$ – Controls somatic maintenance. Variability in maintenance costs will lead to differences in growth.
- $\dot{v}$ – Controls reserve density dynamics. Animals with larger reserve density will have higher weights for the same structural length.
- κ – Controls energy allocation to soma. Animals that allocate a larger fraction to soma will grow faster.

We will experiment with all combinations of one and two parameters and choose the trial parameters based on the fit to the data. The remaining parameters that appear in the growth and feed intake equations will use the values of the breed parameters or the standard parameters if these were not estimated in the breed tier.

In the individual tier, we can also estimate at most two parameters for each individual. The parameters chosen as trial parameters will be the individual parameters in the individual tier, as these values are needed as initial values and pseudo-data values to constrain the estimation. Similarly to the trial tier, the remaining parameters that appear in the growth and feed intake equations will use the values of the breed and standard parameters.

The procedure is schematized in Fig. 3.

3.4. Implementation details

The multi-tier estimation procedure was implemented in Python and made to interface with the DEBtool package for MATLAB. The code package is available at www.github.com/diogo-f-oliveira/DEBtool-Python-Interface.

The developed package allows the independent parameter estimation of statistical universes of the same tier that do not share data (e.g. individuals in the individual tier). It takes as input the files with individual data, a variable describing the tier structure, the tier parameters of each tier, and estimation settings. After the estimation is complete, it outputs a series of files containing the parameters, prediction errors and estimation times for all tiers.

To facilitate the interface with MATLAB, the package uses template estimation files for each estimation tier. These template files contain placeholders for the data and parameters required for the estimation of each statistical universe within a tier. The templates are then populated automatically with the tier parameters and the data collection for the statistical universe, resulting in the creation of the necessary DEBtool estimation files (`mydata_my_pet.m`, `pars_init_my_pet.m`, `predict_my_pet.m`, and `run_my_pet.m`) (Marques et al., 2018). Once these files are constructed, the Python code automatically executes the estimation procedure, retrieves the results, and saves them. This process is then repeated for all statistical universes and all tiers.

For the case study, the estimation was performed on a computer with a Intel(R) Xeon(R) Silver 4108 CPU 1.8 GHz processor and 24 GB of RAM. We used a variation of the classical Nelder–Mead algorithm that implemented restarts every 500 iterations. If the loss function did not improve after a full restart, within a tolerance of 10^{-5}, the estimation was stopped. The results are presented in the following section (Section 4) and estimation times are reported for each tier. A simultaneous estimation, as described in Eq. (4) was also attempted. However, after 48 h it had not converged and the errors were not better than those of the multi-tier estimation, therefore the results were not included in this paper.

The code for the case study is available at www.github.com/diogo-f-oliveira/DEB-Bos-taurus-Mertolenga.

4. Results and discussion

4.1. Breed tier estimation

The parameters for *Bos taurus* Mertolenga obtained are presented in Table 3. The value estimated for the digestion efficiency κ_X was 0.116, which is markedly smaller than the standard value of 0.8. However, since cattle are ruminants, it was expected that the digestion efficiency

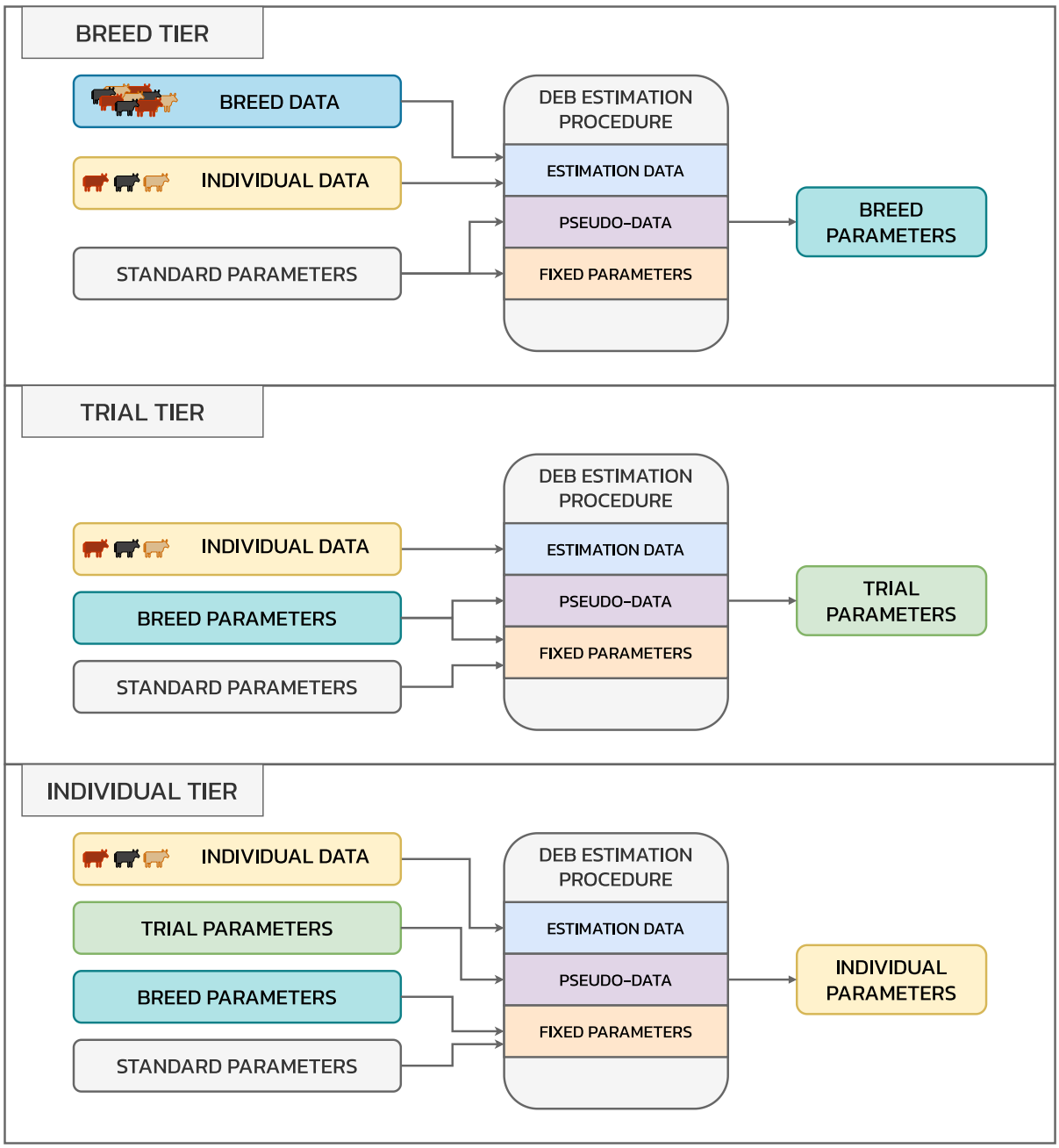

Fig. 3. A schematic of the multi-tier procedure for the case study on Mertolenga cattle.

would be lower, as cellulose digestion is less efficient than animal tissue digestion (Marques et al., 2020). The maximum specific assimilation rate $\{\dot{p}_{Am}\}$ is 13% larger for males than for females. The maturity at weaning is similar for both males and females, however the maturity at puberty is smaller for females as they reach puberty earlier.

The mean relative error (MRE) of the estimation was 6.7%.

Table 1 shows the breed data, the corresponding predictions, and the relative error for each data set. The MRE on breed data was 5.5%. The predictions for most datasets had a relative error below 10%, which is considered acceptable (Marques et al., 2019).

The MRE on individual data was 11.7%, with a MRE of 7.4% on weight data and 16.1% on feed intake data 2. In both datasets, inter-individual variability accounts for a substantial portion of the error, as will be shown through the results of the individual tier estimation (Section 4.3). For feeding data, the individual variability effect is compounded by large scatter in the data, which can be observed in the example feed intake data in Fig. 6).

The estimation converged in 19 min and 50 s. This is the bulk of the total estimation time for any choice of individual parameters. This is explained by a parameter space with a larger dimensionality than the other tiers (15 vs 2), as well as by the need to solve differential equations to compute predictions for some datasets (e.g. weight at birth).

4.2. Trial tier estimation

Table 4 shows the results of the estimation of trial parameters. Each row corresponds to a choice of trial parameters and contains their estimated value for each trial, as well as their relative deviation from the breed parameter. The average errors on weight and feed intake data are also reported.

Table 2
Prediction errors on the individual data for the breed tier estimation.

Individual data	Relative error (± SD)
Weight	7.4 ± 2.2%
Feed intake	16.1 ± 6.5%
All	11.7 ± 3.0%

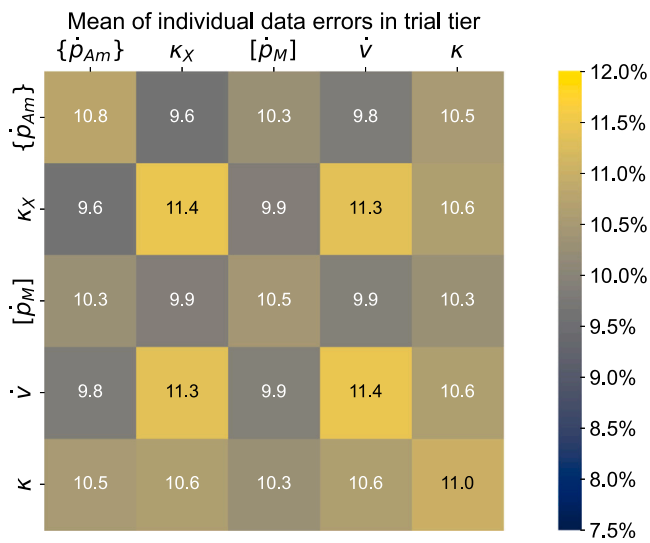

Fig. 4. Mean relative errors on individual data for different choices of trial parameters.

The deviation between the trial parameters and the breed parameters ranges between 0.0% and 12.7%, in absolute terms. The estimations that have κ as a trial parameter result in a parameter value close to or equal to 0.979, which is 5.7% greater than the breed value of 0.934. Upon further inspection, we found that this value is on the limit to meet the condition of reaching puberty, that is, the time to reach puberty is tending to infinity. Moreover, none of the estimations achieve the best fit on weight or feed intake data, which leads to the conclusion that even larger values of κ would be required to achieve a reasonable fit on individual data. Both of these observations point towards κ not being an appropriate choice of trial parameter.

Compared to the error from the breed tier estimation, the introduction of trial parameters in general reduces the error on weight and feed intake data. The effect is more substantial for weight data than for individual feed intake data, where some parameter choices obtain a worse MRE than the breed tier estimation.

Estimations that have $\{\dot{p}_{Am}\}$ or $[\dot{p}_M]$ as a trial parameter generally have lower errors on weight data. Both of these parameters appear in the maximum length L_m and in the von Bertalanffy growth rate $\dot{r}_B$, which means that a change in the parameter values has a considerable effect on the weight curve. We observed that changing these parameters had a larger impact on the prediction for weight than for feed intake, which means that lower deviations from the breed parameters are required to achieve a satisfactory fit. Consequently, due to the influence of the pseudo-data, fitting weight data is prioritized over fitting feed intake data in the minimization of the loss function, which explains the larger MRE on feed intake data of some of the estimations that have these as trial parameters.

For feed intake data, the best trial parameters are κ_X and $\dot{v}$. The digestion efficiency κ_X only appears in the feed intake equation, so any change in its value will only affect that prediction. The energy conductance $\dot{v}$ affects the length $L(t)$ by changing the von Bertalanffy growth rate $\dot{r}_B$ and we observed that increments in the parameter value caused greater changes in the predictions for feed intake than for weight. Similarly to the estimations with $\{\dot{p}_{Am}\}$ or $[\dot{p}_M]$, the estimations with $\dot{v}$ prioritize fitting feed intake data at the expense of the fit on weight data, due to the influence of the pseudo-data.

The estimation errors on individual data for each choice of trial parameters can also be visualized more easily in Fig. 4. Each square in the heatmap corresponds to a choice of trial parameters. The squares on the diagonal show the error of the estimations with a single trial parameter. Darker colors indicate a smaller relative error. It can be observed that combinations that choose one of $\{\dot{p}_{Am}\}$ or $[\dot{p}_M]$ and one of κ_X or $\dot{v}$ have lower estimation errors on individual data. These estimations achieve low prediction errors in both types of individual data because with $\{\dot{p}_{Am}\}$ or $[\dot{p}_M]$ the error on weight data is minimized and with κ_X or $\dot{v}$ the error on feed intake data is minimized.

The trial tier estimation took at most one minute to converge, and on average took only 15 s. Estimations that only had one trial parameters were in general quicker, with the exception of the estimation with κ, which took the most time to converge out of all parameter choices. This is likely due to the proximity of the solution to the region of infeasibility for not reaching puberty.

4.3. Individual tier estimation

Table 5 shows the results of the estimation of individual parameters. Each row corresponds to a choice of individual parameters and contains the mean and standard deviation of each individual parameter, as well as the coefficient of variation of each parameter distribution, that is, the ratio between the standard deviation and the mean of each parameter distribution. The mean and standard deviation of the relative deviation to the trial parameters are also reported, taking into account the trial in which each individual participated. Finally, the mean and standard deviation of the relative error on weight data, feed intake data, and the mean error on individual data are also presented.

In general, the mean deviation in most estimations is close to zero, which means that the individual parameter distribution is centered around the trial parameter values. The mean relative error on individual-level data is once again reduced, particularly for feed intake data. Considering the parameter choices with the lowest relative error on individual data from the breed tier to the trial tier, the mean relative error on feed intake data was reduced from 16.1% to 15.4% and then in the individual tier it was reduced further to 13.2%. In contrast, the mean relative error on weight data was reduced from 8.3% to 3.6% in the trial tier estimation, and the introduction of individual parameters reduced the relative error to 2.1%. This could indicate that the feeding behavior has larger intraspecific variability than growth and that individual parameters are required to properly model the data.

In Fig. 5 we can observe a heatmap of the errors on individual-level data, similar to the one presented in Section 4.2 for the trial tier estimation. Once again, the estimations that have either $\{\dot{p}_{Am}\}$ or $[\dot{p}_M]$ as an individual parameter generally have lower errors on weight data and estimations with κ_X or $\dot{v}$ as an individual parameter generally have lower errors on feed intake data. Estimations with one of each of these parameters ($\{\dot{p}_{Am}\}$ κ_X, $\{\dot{p}_{Am}\}$ $\dot{v}$, $[\dot{p}_M]$ κ_X, and $[\dot{p}_M]$ $\dot{v}$) have lower error on both sets of individual-level data.

The estimation of the individual tier took at most 3 min and 50 s and on average 2 min and 7 s. Regardless of the choice of individual parameters, the estimation time for a single individual took on average 2.4 ± 1.4 s and at most 6.6 s.

In total, the multi-tier estimation took between 20 min 49 s and 24 min 4 s, with an average of 22 min 10 s. The total number of parameters estimated is of 69 for choices with one individual parameters and 123 for choices with two. This parameter space has very large dimensionality, where Nelder–Mead struggles to converge. Nevertheless the multi-tier estimation procedure managed to obtain parameter estimates that achieve low prediction error. This is due to its ability to independently estimate the parameters of each individual,

Table 3
Breed parameters for *Bos taurus* Mertolenga.

Symbol	Core parameter	Value	Units
$\{\dot{p}_{Am}\}$	Surface-specific maximum assimilation rate	4313	$J/d \cdot cm^2$
κ_X	Digestion efficiency	0.116	–
$[\dot{p}_M]$	Volume-specific somatic maintenance rate	106.0	$J/d \cdot cm^3$
$\dot{\upsilon}$	Energy conductance	0.049	cm/d
κ	Allocation fraction to soma	0.926	–
$[E_G]$	Specific cost of structure	7836	J/cm^3
E_H^b	Maturity at birth	7.208×10^6	J
E_H^x	Maturity at weaning	3.925×10^7	J
E_H^p	Maturity at puberty	7.224×10^7	J
$\ddot{h}_a$	Weibull aging acceleration	7.161×10^{-8}	d^{-2}
t_0	Diapause	88	d
δ_M	Shape coefficient	0.221	–
${}_f\{\dot{p}_{Am}\}$	Surface-specific maximum assimilation rate for females	3807	$J/d \cdot cm^2$
${}_fE_H^x$	Maturity at weaning for females	3.701×10^7	J
${}_fE_H^p$	Maturity at puberty for females	5.078×10^7	J

Table 4
Results of the trial tier estimation for different choices of trial parameters. The columns Dev. refer to the relative deviation between the trial parameters and the breed parameter.

Trial	Parameter values					Relative error (± SD) on individual data		
parameters	2020	Dev.	2021	Dev.	Unit	Weight	Feed intake	All
$\{\dot{p}_{Am}\}$	4694	+8.8%	4389	+1.8%	$J/d \cdot cm^2$	**5.3 ± 2.7%**	16.3 ± 6.8%	**10.8 ± 3.2%**
κ_X	0.110	−5.0%	0.122	+4.9%	–	7.4 ± 2.2%	**15.4 ± 5.5%**	**11.4 ± 2.5%**
$[\dot{p}_M]$	93.5	−11.8%	98.2	−7.4%	$J/d \cdot cm^3$	**4.5 ± 2.2%**	16.4 ± 6.9%	**10.5 ± 3.3%**
$\dot{\upsilon}$	0.053	+6.4%	0.046	−7.6%	cm/d	**7.4 ± 2.1%**	**15.4 ± 5.4%**	**11.4 ± 2.5%**
κ	0.979	+5.7%	0.979	+5.7%	–	**5.7 ± 2.1%**	16.3 ± 6.8%	**11.0 ± 3.2%**
$\{\dot{p}_{Am}\}$ κ_X	4711 0.117	+9.2% +0.5%	4738 0.129	+9.9% +11.3%	$J/d \cdot cm^2$ –	**3.6 ± 1.9%**	**15.7 ± 5.6%**	**9.6 ± 2.8%**
$\{\dot{p}_{Am}\}$ $[\dot{p}_M]$	4609 101.5	+6.9% −4.3%	4167 94.1	−3.4% −11.2%	$J/d \cdot cm^2$ $J/d \cdot cm^3$	**4.4 ± 2.3%**	16.2 ± 6.6%	**10.3 ± 3.1%**
$\{\dot{p}_{Am}\}$ $\dot{\upsilon}$	4710 0.049	+9.2% −0.8%	4641 0.043	+7.6% −12.7%	$J/d \cdot cm^2$ cm/d	**3.9 ± 2.0%**	**15.8 ± 5.8%**	**9.8 ± 2.8%**
$\{\dot{p}_{Am}\}$ κ	4606 0.969	+6.8% +4.6%	4285 0.978	−0.7% +5.6%	$J/d \cdot cm^2$ –	**4.7 ± 2.6%**	16.3 ± 6.7%	**10.5 ± 3.2%**
κ_X $[\dot{p}_M]$	0.113 95.4	−3.0% −10.0%	0.125 93.8	+7.4% −11.6%	– $J/d \cdot cm^3$	**4.1 ± 1.9%**	**15.6 ± 5.5%**	**9.9 ± 2.7%**
κ_X $\dot{\upsilon}$	0.110 0.049	−5.4% −0.8%	0.118 0.047	+1.8% −5.5%	– cm/d	**7.3 ± 2.2%**	**15.4 ± 5.4%**	**11.3 ± 2.5%**
κ_X κ	0.112 0.979	−3.9% +5.7%	0.123 0.979	+6.1% +5.7%	– –	**5.7 ± 2.1%**	**15.5 ± 5.5%**	**10.6 ± 2.6%**
$[\dot{p}_M]$ $\dot{\upsilon}$	94.8 0.051	−10.6% +4.1%	94.9 0.045	−10.5% −9.5%	$J/d \cdot cm^3$ cm/d	**4.1 ± 2.0%**	**15.7 ± 5.7%**	**9.9 ± 2.8%**
$[\dot{p}_M]$ κ	97.3 0.982	−8.3% +6.0%	101.4 0.969	−4.3% +4.6%	$J/d \cdot cm^3$ –	**4.0 ± 2.2%**	16.5 ± 7.0%	**10.3 ± 3.3%**
$\dot{\upsilon}$ κ	0.052 0.979	+5.3% +5.7%	0.045 0.979	−8.5% +5.7%	cm/d –	**5.7 ± 2.1%**	**15.5 ± 5.6%**	**10.6 ± 2.6%**

thereby partitioning the parameter space into smaller, easier to solve parameter spaces. As a result, the computational effort required is drastically reduced.

4.4. Choosing individual parameters

In the modeling of Mertolenga cattle, *ad libitum* feeding was assumed, which means we have a constant scaled functional response $f = 1$. Under this assumption, both weight and feed intake are functions of the structural length L which follows von Bertalanffy growth and tends to the maximum size $L_m = \kappa\{\dot{p}_{Am}\}/[\dot{p}_M]$. Therefore, the simplest explanation for the observed variability in the weight and feed intake data would be an intraspecific change in maximum size $L_m = \kappa\{\dot{p}_{Am}\}/[\dot{p}_M]$ by virtue of a single parameter: κ, $\{\dot{p}_{Am}\}$ or $[\dot{p}_M]$. However, choosing only one of these parameters as an individual parameter leads to larger errors on individual data.

Since the duration of the trials is small compared to the lifespan of the species, another explanation would be an intraspecific change in the von Bertalanffy growth rate $\dot{r}_B$, which is a function of the parameters mentioned previously and the energy conductance $\dot{\upsilon}$. Although choosing $\dot{\upsilon}$ as an individual parameter leads to improved predictions for feed intake, it comes at the cost of the prediction for weight data.

Fig. 6 shows weight and feed intake data for an individual and the corresponding DEB model prediction under the choice of $\{\dot{p}_{Am}\}$ as an individual parameter. With just one individual parameter, the DEB model cannot fit properly to the data. The minimization led to

Table 5
Results of the individual tier estimation for different choices of individual parameters. The column Deviation gives the mean and the standard deviation (SD) of the relative deviation between the individual parameters and the trial parameters, taking into account the trial that each individuals belongs to.

Individual	Parameter values			Deviation	Relative error (± SD) on individual data		
parameters	Mean ± SD	CV	Unit	Mean ± SD	Weight	Feed intake	All
$\{\dot{p}_{Am}\}$	4580 ± 434	0.09	$J/d \cdot cm^2$	+1.0 ± 8.8%	**5.1 ± 2.9%**	**14.5 ± 3.4%**	**9.8 ± 2.8%**
κ_X	0.117 ± 0.013	0.11	–	+0.5 ± 9.4%	7.4 ± 2.2%	**13.2 ± 2.4%**	**10.3 ± 1.0%**
$[\dot{p}_M]$	93.3 ± 10.5	0.11	$J/d \cdot cm^3$	−2.8 ± 10.4%	**3.6 ± 1.8%**	**15.8 ± 5.6%**	**9.7 ± 3.4%**
$\dot{v}$	0.049 ± 0.007	0.15	cm/d	+0.3 ± 11.9%	7.5 ± 2.5%	**13.3 ± 2.4%**	**10.4 ± 1.2%**
κ	0.979 ± 0.000	0.00	–	+0.0 ± 0.0%	**5.7 ± 2.1%**	16.3 ± 6.8%	**11.0 ± 3.2%**
$\{\dot{p}_{Am}\}$ κ_X	4857 ± 245 0.126 ± 0.012	0.05 0.09	$J/d \cdot cm^2$ –	+2.8 ± 5.1% +2.5 ± 7.1%	**2.1 ± 0.7%**	**13.5 ± 2.5%**	**7.8 ± 1.4%**
$\{\dot{p}_{Am}\}$ $[\dot{p}_M]$	4317 ± 531 92.8 ± 7.8	0.12 0.08	$J/d \cdot cm^2$ $J/d \cdot cm^3$	−1.5 ± 10.0% −5.1 ± 5.0%	**3.7 ± 2.1%**	**14.1 ± 3.0%**	**8.9 ± 2.2%**
$\{\dot{p}_{Am}\}$ $\dot{v}$	4808 ± 288 0.045 ± 0.005	0.06 0.12	$J/d \cdot cm^2$ cm/d	+2.9 ± 6.2% −3.3 ± 7.9%	**2.5 ± 0.8%**	**13.5 ± 2.5%**	**8.0 ± 1.4%**
$\{\dot{p}_{Am}\}$ κ	4542 ± 340 0.975 ± 0.012	0.07 0.01	$J/d \cdot cm^2$ –	+2.4 ± 7.2% +0.1 ± 1.0%	**3.8 ± 2.0%**	**15.5 ± 6.7%**	**9.7 ± 3.4%**
κ_X $[\dot{p}_M]$	0.121 ± 0.013 90.5 ± 5.3	0.10 0.06	– $J/d \cdot cm^3$	+1.4 ± 8.7% −4.3 ± 5.5%	**2.6 ± 0.9%**	**13.5 ± 2.5%**	**8.0 ± 1.3%**
κ_X $\dot{v}$	0.113 ± 0.010 0.046 ± 0.002	0.09 0.04	– cm/d	−1.4 ± 7.6% −2.9 ± 2.7%	**7.2 ± 2.2%**	**13.2 ± 2.4%**	**10.2 ± 1.1%**
κ_X κ	0.118 ± 0.013 0.979 ± 0.000	0.11 0.00	– –	+0.6 ± 9.4% +0.0 ± 0.0%	**5.7 ± 2.1%**	**13.3 ± 2.5%**	**9.5 ± 1.1%**
$[\dot{p}_M]$ $\dot{v}$	90.7 ± 6.0 0.047 ± 0.007	0.07 0.14	$J/d \cdot cm^3$ cm/d	−4.3 ± 6.3% −1.1 ± 11.0%	**2.7 ± 0.8%**	**13.5 ± 2.5%**	**8.1 ± 1.3%**
$[\dot{p}_M]$ κ	98.4 ± 5.9 0.970 ± 0.032	0.06 0.03	$J/d \cdot cm^3$ –	−1.0 ± 5.8% −0.5 ± 3.1%	**3.8 ± 2.0%**	**16.0 ± 5.5%**	**9.9 ± 3.4%**
$\dot{v}$ κ	0.049 ± 0.007 0.979 ± 0.000	0.15 0.00	cm/d –	+0.2 ± 12.1% +0.0 ± 0.0%	5.7 ± 2.3%	**13.4 ± 2.5%**	**9.6 ± 1.2%**

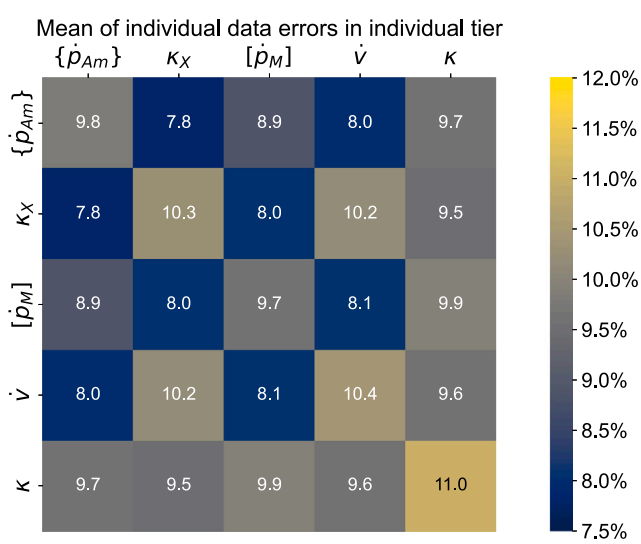

Fig. 5. Mean relative errors on individual data for different choices of individual parameters.

a compromise, fitting neither dataset well. The value of $\{\dot{p}_{Am}\}$ should increase to reduce the weight error, but this would lead to a larger feed intake, which will increase the error, as the prediction is already above what would be the mean line through the data. Similar trade-offs occur when $[\dot{p}_M]$, κ or $\dot{v}$ are the only individual parameters.

These observations point to the need for at least two parameters in order to fit the model properly and avoid minimization trade-offs. Of the four combinations of individual parameters with the least error, $\{\dot{p}_{Am}\}$ and κ_X are the most biologically reasonable to vary. The variability in weight and feed intake data can be attributed to changes in size and ingestion characteristics.

The surface-specific maximum assimilation rate $\{\dot{p}_{Am}\}$ is directly related to ingestion and has an influence in the growth and the maximum size of organisms. It is typically used to model changes in size, particularly between males and females by proxy of the zoom factor z (Kooijman, 1986). Moreover, it is diet dependent and can model changes in feed net energy content.

The digestion efficiency κ_X is the parameter, of the five hypothesized, that is strictly related to feeding, in the sense that it is the only one that does not appear in the growth equation. Furthermore, feed efficiency is a subject extensively studied in livestock production. It is typically considered to be independent of weight, measured in identical conditions for all animals, and has been demonstrated to have an intraspecific component (Kennedy et al., 1993; Herd and Arthur, 2009). The parameter most similar in concept is the digestion efficiency, as it does not have an influence on weight gain. It is also related to the concepts of net energy fraction and digestibility, together with the defecation efficiency κ_P, which has also been shown to have a intraspecific component (Tharangani et al., 2020; Cabezas-Garcia et al., 2017).

Moreover, the estimation with $\{\dot{p}_{Am}\}$ and κ_X as individual parameters has the lowest error on weight data (2.1%) and comparable error on feed intake data (13.5%) to the lowest error (13.2% with κ_X and $\dot{v}$). Overall, this choice of individual parameters leads to the lowest MRE on individual data (7.8%). Finally, the minimization trade-off disappears and the model produces accurate predictions, as can be seen in Fig. 7.

Fig. 8 shows the distributions of $\{\dot{p}_{Am}\}$ and κ_X as well as a scatterplot of the parameter values, along with the estimates for the breed and trial tiers. On average, the individual parameter values of $\{\dot{p}_{Am}\}$ and κ_X 11.9% and 8.8% larger than the breed parameter values, respectively. These differences reflect changes in environmental conditions, which

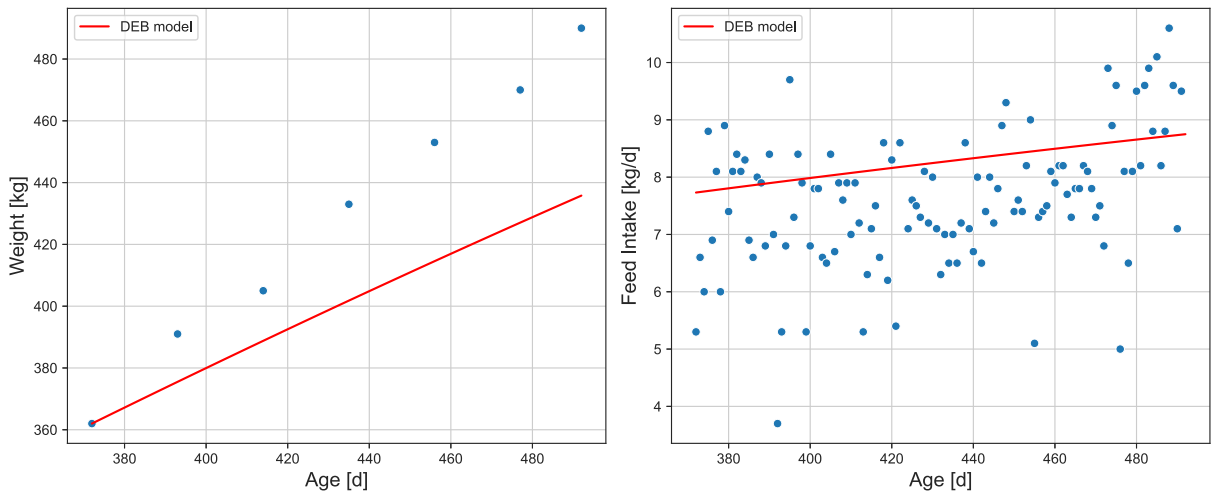

Fig. 6. Example of the fit of the DEB model (red) when $\{\dot{p}_{Am}\}$ is an individual parameter. Weight on the left and feed intake on the right.

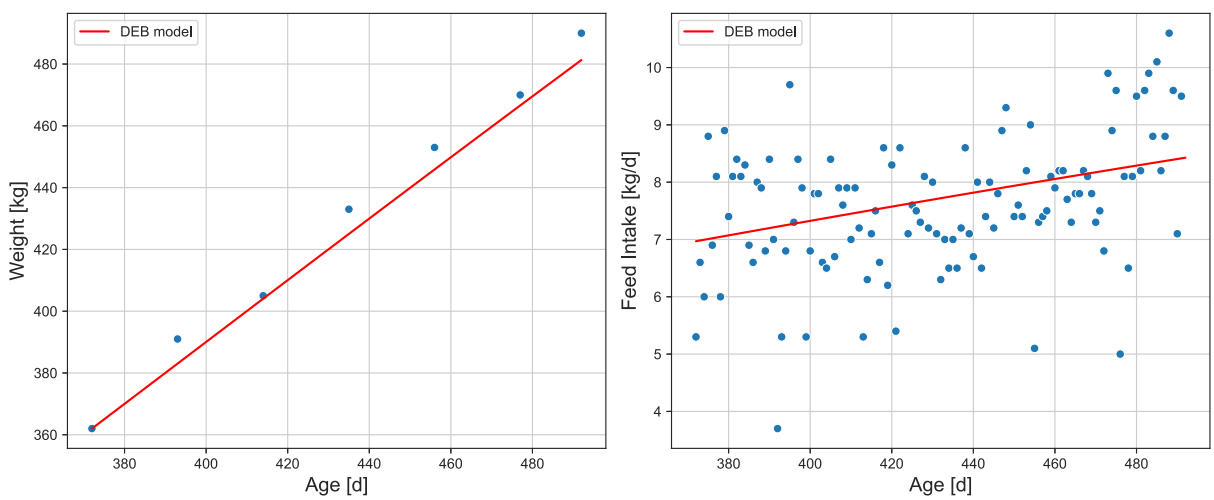

Fig. 7. Prediction of the DEB model (red) for the same individual of Fig. 6 when $\{\dot{p}_{Am}\}$ and κ_X are individual parameters. Weight on the left and feed intake on the right.

can likely be attributed to the diets fed to the animals. Higher values of $\{\dot{p}_{Am}\}$ suggest higher net energy content, whilst higher values of κ_X indicate that the animals can extract more energy from the diet.

Comparing the trial parameters of each year, we can find evidence that there were differences in the diets, despite the effort to have the same nutritional content. For the same structural length, the similar values of $\{\dot{p}_{Am}\}$ indicate similar net energy values for both diets; however, the differences in κ_X imply that the net energy fraction was not equal. This observation is supported by the lack of overlap between the clusters of individual parameters of each trial.

5. Conclusions and future work

In this paper, we have presented a multi-tier methodology for estimating individual-specific parameters of Dynamic Energy Budget models. Individual parameters can only be accurately estimated from individual data; however, it is not always practical to collect sufficiently complete data to accurately estimate most core DEB model parameters at the individual level. Consequently, it is necessary to complement individual data with more general data on groups of individuals, such as species or populations. This data can bias the estimation of individual parameters if the organisms are not similar or if they experienced different environmental conditions compared to the individuals in the estimation. On the other hand, prioritizing individual data will lead to parameter distributions with high variance, where the parameters of some individuals may not be biologically reasonable.

Our proposed methodology addresses this trade-off by means of a hierarchical tier structure. By organizing the data into tiers, we identify the possible sources of bias and structure the estimation to use data from similar organisms in similar conditions. In each tier, we estimate only a subset of parameters, which are then used as pseudo-data to anchor the estimation of individual parameters. Each new tier estimate produces a pseudo-data value with lower bias with respect to the individual parameters. Moreover, by fixing most parameters and regularizing the estimation, we obtain low-variance distributions with biologically sensible individual parameter values.

The multi-tier methodology is built on top of the methods and principles of the standard DEB estimation procedure. First, it shares the same loss function and minimization procedure to find parameter values. Second, it makes use of pseudo-data to regularize the estimation thereby avoiding unrealistic parameter estimates. Third, and most important, it follows the estimation-in-context principles and biological reasonableness is one of its key pillars. Moreover, the tier structure enables fast estimation and iteration and avoids high-dimensional parameter spaces where convergence issues can arise.

We applied our methodology to a case study on Mertolenga cattle. Three tiers were considered: breed, trial, and individual. Due to the characteristics of the individual weight and feed intake data, at most

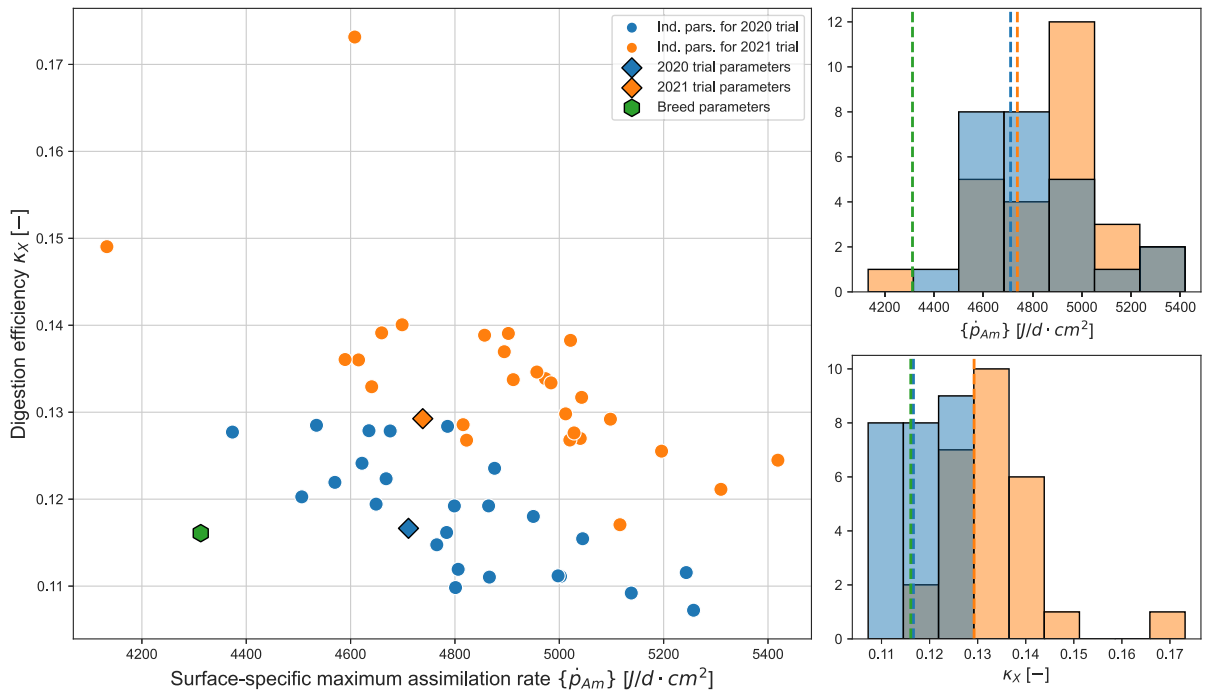

Fig. 8. Scatterplot and histogram showing the individual parameter values of $\{\dot{p}_{Am}\}$ and κ_X. Individuals from the 2020 trial are colored blue, while those from the 2021 trial are colored orange. For reference, the trial parameter values are marked with diamonds in the respective colors and the breed parameter values with hexagons in green.

two trial and individual parameters could be estimated. We tested several combinations of parameters and analyzed the results. We concluded that at least two individual parameters were needed to properly fit both datasets and that $\{\dot{p}_{Am}\}$ and κ_X are the most appropriate choice, both in terms of prediction errors and biological reasonableness. The analysis of the trial parameters and individual parameter distributions uncovered differences in the net energy fraction of the diets of each performance trial.

For future work, we see the need to explore other ways to anchor the estimation of individual parameters. Grounding the estimation will always be required, however using pseudo-data is not the only alternative. For example, if data on the distribution of a given trait can be collected, then it could be included in the estimation by tuning the weight of the dataset so as to ensure the prediction is within the dispersion of the distribution. If confidence intervals can be determined for tier parameter estimates, then these could be used to constrain the estimation of the next tier. Both Bayesian approaches (Johnson et al., 2013; Boersch-Supan and Johnson, 2019) and covariation-based approaches (Lika and Kooijman, 2024; Marques et al., 2019) offer methods to obtain parameter confidence intervals. Finally, uncovering patterns in the distributions of individual parameters by using non-parametric estimation methods, such as ours, could also lead to the development of more appropriate priors and regularization techniques.

Another avenue of research is to apply techniques from the machine learning literature. In essence, the choice of individual parameters is similar to that of model selection. Techniques, such as cross-validation or bootstrapping, and criteria that take into account model complexity, e.g. the Bayesian Information Criterion (BIC) or the Alkaike Information Criterion (AIC), could be integrated into the methodology and aid in the selection of individual parameters.

CRediT authorship contribution statement

Diogo F. Oliveira: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Gonçalo M. Marques:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Nuno Carolino:** Writing – review & editing, Investigation, Funding acquisition, Data curation. **José Pais:** Writing – review & editing, Investigation, Funding acquisition, Data curation. **João M.C. Sousa:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Tiago Domingos:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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